

Structure

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12.0 INTRODUCTION

The data is stored and organized through file structure. Software applications and platforms store the data in files. Operating systems manage critical configuration through file and even databases also store the data in files such as transaction logs. Files essentially serves as the key building blocks for data organization. We can enforce governance such as data integrity checks, security rules, access controls, sharing controls for files.

In this unit we discuss various aspects of file such as file organization, file types and related concepts.

12.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of files,
- understand the file organization concepts,
- understand sequential files and direct file organization,
- Understand indexed file organization and indexed sequential files

12.2 BASIC CONCEPTS OF FILE

A file is a collection of data records. The key operations of the file are as follows:

- File creation when the file gets created initially.
- File update when we update the file with new data records or modify existing data records.

- File deletion when we delete the file
- File merging that combines the content of multiple files into a single file.
- File searching that involves searching for a specific data record in a file.

A data record in a file encapsulates the data of a single entity. For instance, an employee data record captures details of a single employee such as employee id, employee name, employee address, employee phone number and employee date of join.

Each data record is uniquely identified by a key. For instance, in the employee data record, employee id is the key.

The table 1 provides the employee data records in an organization:

Employee Number	Employee Name	Employee Address	Department	Date of Join
45278	Employee 1	Kanpur	Sales	01-Jan-2020
167352	Employee 2	Mysore	Development	01-Mar-2020

12.3 FILE ORGANIZATION

File organization defines the way the data record is stored and retrieved from the files. File organization provides the data record relationship specifying how the data records are mapped to the disk blocks. We can either store fixed length record in the files or use flexible file structure to support data records of varying length.

There are multiple ways of file organization such as sequential, direct, indexed sequential and others. We can select the file organization type based on the use case, information retrieval performance, ease of information management and others.

Given below are the most commonly used file organizations:

- Sequential files – The data records are stored in a pre-defined order. For instance, the data records are sequenced based on the key values. In this file organization type we need to sequentially access the record.
- Relative files – The data record is stored at a fixed location that can be accessed using a static key value. As we can access the data record using a fixed value, we can retrieve or insert the data record randomly in this file organization type.
- Direct files – Similar to the relative file that supports non-integer key value.
- Indexed sequential files – In this file organization we add an index to provide the random access to the sequential files.
- Indexed files – The data records are indexed to improve the performance of the data retrieval.

Figure 1 provides various file organizations that are most commonly used.

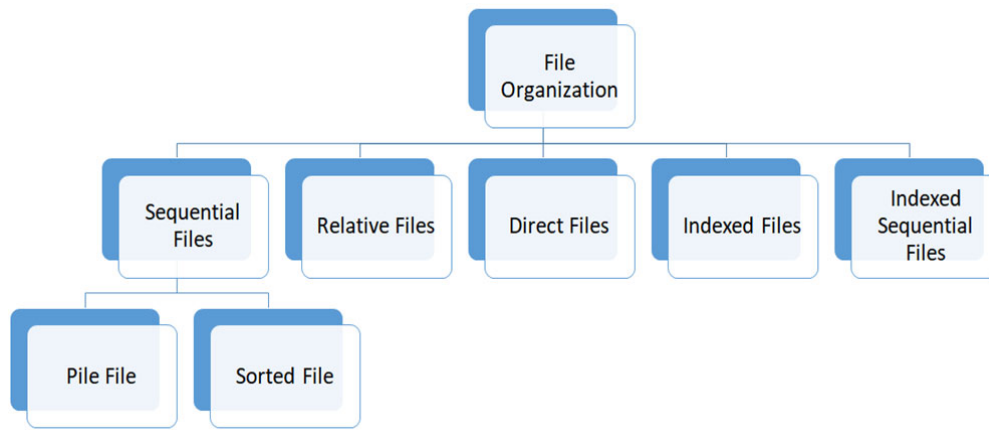


Figure 1 Types of File organizations

12.3.1 Key drivers for file organization

Given below are the main drivers for the file organization:

- File organization helps in optimal design of file records so that they can be retrieved as quickly as possible.
- Improve the performance of file operations such as insert, update and delete.
- Ensure the data consistency and avoid duplicates during file operations.
- Provide optimal storage for managing the files.

12.4 SEQUENTIAL FILES

We write the data records in a specific sequence in sequential files. The data is also accessed in the order it is written to the physical device. We usually order the data records based on the key value. In Sequential files the reading happens from the beginning of the file and write happens at the end of the file.

The legacy storage systems such as tape storage use sequential files.

12.4.1 Operations on sequential files

We discuss the main operations on the sequential files:

Insert operation

As the sequential files are organized based on their keys, when we insert we should find the exact position of the new key value and insert the key over there.

If there are multiple insertions, we can batch them to optimize the performance optimization. We can collect all the new inserts into transaction log and then sort the new keys in order in the transaction log. We can then merge the entries in the transaction log into the main file in batch mode.

Delete Operation

In the delete operation, we need to reclaim the space from the deleted record. Here as well we can use the concept of transaction log for performance optimization. We can add the delete marker for the key position in the transaction log and later we can merge the changes to the main file in batch mode. During merge operation, we drop the data records in the main file based on the marker.

Update Operation

For performing the update operation, we need to add the delete marker for the corresponding key and then insert the new data record for the corresponding key in the transaction log. Like always we merge the transaction log to the main file

Retrieval Operation

We read the data record based on its key. For stale data reads, we can directly read the data record for the key from the main file. To maintain read consistency, we need to check if there are any pending updates in the transaction log and if so we need to merge the changes (deletion/insertion).

12.4.2 Types of sequential files

There are mainly two types of sequential files – Pile files and sorted files.

Pile Files

In Pile files, the data records are inserted in the order of their arrival. For instance, let's consider the below given incoming Data records –

DR1, DR3, DR5, DR2

The data records are inserted as given in Figure 2:

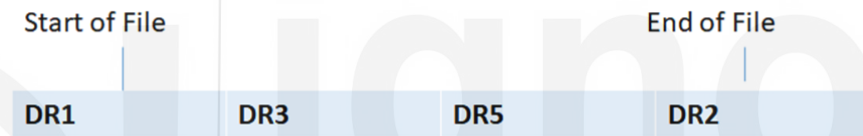


Figure 2 Pile File

For searching the records, the file is searched from the beginning till the desired record is found. For deleting a record, we search for the record and add a deletion marker for the found record.

For inserting a new record, it will be done at the end of the file. The new record DR4 is to be inserted as shown in Figure 3



Figure 3 Pile File Insertion

Sort Files

In case of sort files, the data records are sorted based on the key and are inserted into the file.

For instance, let's consider the below given incoming Data records –

DR1, DR3, DR5, DR2

The data records are inserted as given in Figure 4:

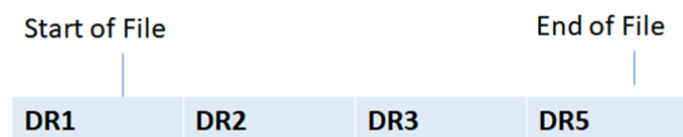


Figure 4 Sort File

While inserting a new record the key of the new record is sorted along with the existing records and then inserted. For instance when we try to insert the data record DR4, the sorted insert is depicted in Figure 5



Figure 5 Sort file insert

12.4.3 Advantages and disadvantages of sequential files

In this section we discuss the main advantages and disadvantages of the sequential files.

Advantages

The main advantages of sequential files are as follows:

1. Simple and easy to implement
2. Provides best usage of the storage space.
3. Can be used for data archival needs.
4. The underlying storage systems provide low-cost storage option.
5. Can be stored in inexpensive device like tape device.

Disadvantages

The main disadvantages of the sequential files are as follows:

1. The update operation is costlier and time consuming.
2. The access speed is slow due to the sequential access.
3. It is not possible to directly access a given data record key
4. All records should have uniform structure.
5. The insert time of sorted file is high.
6. Overall processing is slow.

12.4.4 Use cases for sequential files

We generally use sequential files for long-term archival of data and for batch processing operations. We also use sequential access for storing large amount of data that tolerates slow retrieval time.

Check Your Progress – 1

1. A _____ in a file encapsulates the data of a single entity
2. Each data record is uniquely identified by a _____.
3. In _____ files, the data records are inserted in the order of their arrival.
4. For optimizing the performance, we get all the updates into a _____ and later merge it into main file.
5. The two types of sequential files are _____ and _____.
6. Define a data record for a typical university's student enrolment system.

12.5 DIRECT FILE ORGANIZATION

In the direct file organization, we directly access the file by its key. To enable this direct access, we need to map the key to the address where the data record is stored. Direct access is also known as random access. The file records are stored in direct access storage device (DASD) like hard disk, CD, magnetic disk. We randomly place the record throughout the hard disk.

We have depicted the direct file organization in Figure 6

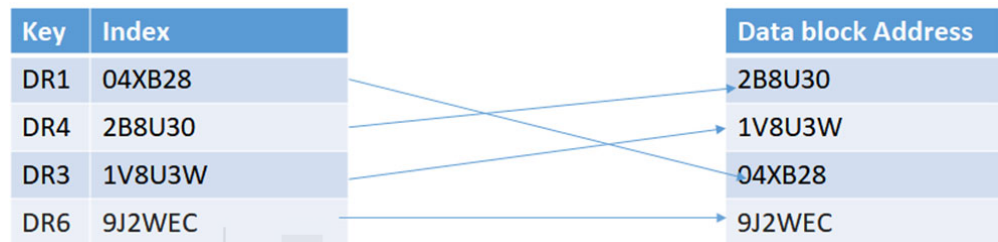


Figure 6 Direct File Organization

We use hash function to convert the key into the address. Naturally direct file access is faster compared to sequential files.

In direct file organization, the records are stored at a known addressed as depicted in Figure 6.

12.5.1 Hash Function and Collision Resolution

The most popular hash function is the modulo based function that computes the remainder value as the hash value based on the key. The hash value is then used to store the data record.

When there are huge set of keys, multiple keys will end up with same hash value leading to collision. In such cases, we use the collision resolution technique. The separate chaining technique chains the values to the same slot and open addressing technique probes for next available slot for placing the value.

12.5.2 Advantages and disadvantages of Direct file organization

We shall look at the main advantages and disadvantages of the direct file organization.

Advantages

Given below are the main advantages of direct file organization:

1. As we can directly access the data, the retrieval is fast that helps in transaction management systems such as relational database systems.
2. The insert, update and delete operations are faster.
3. The direct file organization is efficient in storage and retrieval of large data.
4. Key-based search is faster.
5. Can be used for real-time transactions that needs optimal performance.
6. Concurrent processing is possible

Disadvantages

Given below are the main disadvantages of direct file organization:

1. The required storage technology is costlier.
2. The usage of storage space is sub optimal.
3. Insert, delete and update needs update of the index table.

12.6 INDEXED SEQUENTIAL FILE ORGANIZATION

Indexed sequential file organization supports direct access of keys and also sequential access by keys.

We use the indexed sequential file organization for hierarchical data organization. For instance, let's consider a use case for retrieving the data record for the states. We use the indexed/direct file for getting the country. The country data record points to the sequential file storing the records of its constituent states. Indexed sequential files can be stored only on random access device such as magnetic disk.

To implement the indexed sequential file, we decouple the index and data files. The index file is structured hierarchically in tree structure whereas the data file stores the information sequentially.

We use two types of indexes static index and dynamic index.

We have depicted the indexed sequential file in Figure 7.

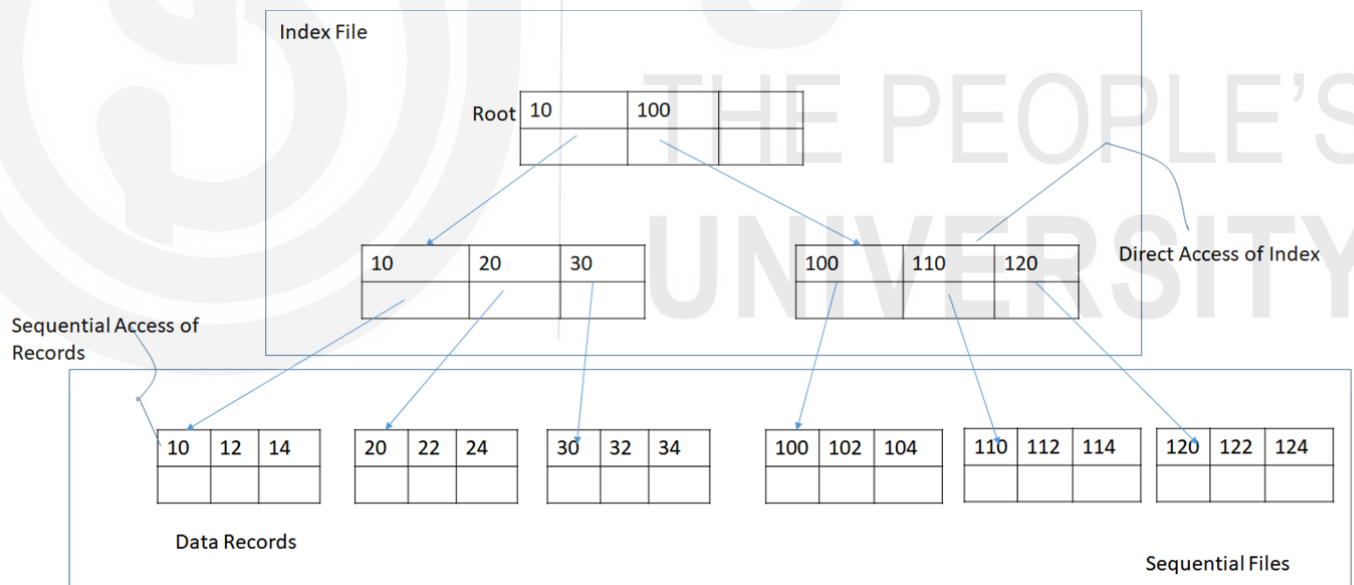


Figure 7 Indexed sequential file

As depicted in Figure 7, the indexed sequential file is depicted as a two-level hierarchy. The first level holds the index to each of the three records in the data file. The second level holds the sequential set of records.

The first level is accessed directly where we get the keys from the index file. In the second level we store the data record sequentially. As the data records are stored sequentially, the key in the first level just points to the first record in the second level from where it can be accessed sequentially.

When we have to insert the record, we update the data file based on the sequence and update the index accordingly. Static index and dynamic index are two main types of indexes. In static indexes the update to the records in the data file does not change the index structure whereas in the dynamic index, the updates to the data file changes the index structure.

12.6.1 Advantages and disadvantages of Indexed sequential file organization

We shall look at the main advantages and disadvantages of the indexed sequential file organization

Advantages

Given below are the main advantages of indexed sequential file organization:

- As the records are directly accessed, the access speed is high
- Record insert is very fast.

Disadvantages

Given below are the main disadvantages of indexed sequential file organization:

- Usage of storage space is sub optimal
- The implementation is costly due to the required of costly hardware
- Extra storage for index file is required.

Check Your Progress – 2

1. In the direct file organization, we directly access the file by its ____
2. The function that converts the key to address is called ____
3. The first level in the indexed sequential access is done ____ and second level is accessed ____
4. The data records in direct file organization are stored in ____ storage device
5. ____ file organization for managing hierarchical data

12.7 SUMMARY

In this unit we mainly discussed about file organization and various types of file organization. A file is a collection of data records. The key operations of the file are File creation, file update, file deletion, file merging, file searching. File organization defines the way the data record is stored and retrieved from the files. We write the data records in a specific sequence in sequential files. There are mainly two types of sequential files - Pile Files and sort files. In Pile files, the data records are inserted in the order of their arrival. In case of sort files, the data records are sorted based on the key and are inserted into the file. In the direct file organization, we directly access the file by its key. Indexed sequential file organization supports direct access of keys and also sequential access by keys.

12.8 SOLUTIONS/ANSWERS

Check Your Progress – 1

1. Data record.

2. Key
3. Pile files
4. Transaction log
5. Pile files and sort files
6. A sample student enrollment structure is as follows:
Student Number, Student Name, Date of Enrollment, Address, Programme

Check Your Progress – 2

1. Key
2. Hash function
3. Directly and sequentially
4. Direct access
5. Indexed sequential

12.9 FURTHER READINGS

References

https://en.wikipedia.org/wiki/File_system
https://en.wikipedia.org/wiki/Sequential_access
https://en.wikipedia.org/wiki/Indexed_file
<https://en.wikipedia.org/wiki/ISAM>